

Project 1

Technical Review

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Project 1: Technical Review

- Group based project
 - Each group has up to 4 members
 - You can form a group with fewer than 4 members.
 - The assessment criteria will be the same for any group size.
- Grouping:
 - please edit the online form below to create your group.
 - https://docs.google.com/spreadsheets/d/1hSPnlG_aP9eG4eJwBiPBYoeyHkTDE7Ik3Peg1rwIJXM/edit?usp=sharing
 - **Please enter your full name exactly as shown on your student card (important!)**
 - The first person for each group is the coordinator.

- Please complete the grouping by Mon Week 3
 - **Everyone should enter your name in the online form.**
 - If you do not join any existing group, please create a new group and enter your name. We will merge small groups if necessary.
 - If your group is not full (fewer than 4 members), additional members may be added when we finalize the groups.
 - The final grouping will be released in Week 3. You need to verify your grouping.

Submission

- Please submit 3 files using the following naming guidelines:
 - 1 “group_XX_report.pdf”
 - Your main report PDF, XX is your group ID.
 - 2. “group_XX_contribution.pdf”
 - A PDF describes the contribution for each member
 - 3. “group_XX_code.zip”
 - A ZIP package for the source code
 - no datasets, no models, no external links
- Where to submit
 - Please submit your zip file in NTULearn:
the course site -> Assignments -> submission link for project 1

Submission

- Submission requirement
 - One group only makes one submission.
 - You can submit multiple times (no limit).
 - You don't need to provide comments in your code.
 - You don't need to submit your **datasets or models**.
 - **Should not provide external links!**
 - cloud-hosted files are not accepted!
- Submission due date: end of week 9 (18 Oct 2026)

Requirements for the technical review report

1. Select a topic/task from the list below:

- Clustering
 - Link Analysis
 - Similarity Search
 - Graph neural networks or deep networks on graphs
 - Graph clustering or graph community detection
 - Recommender System
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- Any sub-topics that belong to the above topics
 - Any topics/applications that use the above techniques

- 2. Select methods for review and comparison
 - Select at least 2 methods (algorithms)
 - You can choose any methods
 - They can be any methods published online, including but not limited to what you learn in the course
E.g., K-means and K-means++, or K-mean and DBSCAN
 - You can use any public code
 - You can propose your own methods
 - e.g., extensions of existing methods
 - reviewing more than 2 methods is encouraged
 - You are encouraged to explore methods that are not covered in the course.

3. Use at least 2 datasets

- Any datasets, including public datasets or the dataset you create.
- No requirement on the size/scale of the dataset
- You can use more than 2 datasets

4. Experimental analysis and comparisons

- Should not copy results from others (including published papers)
 - You need to produce the results by yourself.
- Analyze and discuss the strengths and weaknesses of the methods
- Analyze and discuss the parameter settings
 - E.g., the number of K in K -means
- Ablation study of the key components in a method (if applicable)
 - Only applicable to the methods that have multiple components

4. Experimental analysis and comparisons

(continued)

- Discuss the key factors that affect the performance
 - E.g., K-means are sensitive to centroid initialization
- Compare the performance of the methods and provide discussions
- Illustrate and discuss the successful cases and failure cases (if applicable)
- Other possible analysis

5. Format

- At least 4 pages, no upper limit.
(font-size: 11pt or 12pt)
- Include the following sections:
Abstract/Introduction/Methods/Experiments/Conclusion

6. Rubrics

A good report should meet the following:

- Meet the requirements well.
- Well organized and structured
- Clear explanation of the tasks and the methods.
- Comprehensive and in-depth analysis.
- Present clear and insightful comparisons and discussions
- Good visualizations for analysis.
- Target challenging datasets/applications/methods.
- Discuss or implement possible solutions to improve the methods

About teamwork

1. Ensure all members work together and everyone is involved and contributes to the project.
2. A student with minimal contribution may receive a significant penalty in marks.
3. Sufficient coordination and communication in the team.
 - If a team member fails to contribute due to insufficient communication and coordination, the entire team will face penalties.
4. Report an issue
 - If a team member refuses to contribute, please email the instructor at least 4 weeks before the submission deadline.
 - If you cannot report the issue 4 weeks before the deadline, it indicates that the entire team lacks communication and coordination. We will not handle any issues if they are not reported in time.

Contact:

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Or send me message in MS Teams